

Heuristic Evaluation

Structure of the individual report

Part I: Your Name

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Part II: Project Description

The project I evaluated is called TrailSight, and I reviewed the Smartwatch Prototype. TrailSight helps hikers make safer decisions by showing community-added, geotagged tips such as photos, warnings, and short notes at confusing points on a trail.

The smartwatch prototype consists of several circular paper screens representing the interface: Home, Routes, Route Details, Tip List, Tip Details, Add Tip, and Confirmation. Navigation is shown through arrows around the watch frame to simulate taps or swipes.

The evaluation focused on three tasks provided by the group:

- Simple: Check tips for a known tricky spot before starting a hike.
- Moderate: View tips at a fork during the hike and choose the correct path.
- Complex: Add a new tip with a description and importance level.

Part III: Evaluation Execution

I conducted the heuristic evaluation in the HCI lab during the scheduled session last week. I approached the group responsible for the TrailSight smartwatch prototype, and they briefly introduced their concept, the purpose of the system, and the three tasks I was expected to perform. They provided the physical paper prototype, which consisted of several smartwatch-shaped screens and small arrows representing navigation actions.

One group member acted as the “computer” by changing the paper screens according to my interactions, while another acted as the facilitator. I performed each of the three tasks step by step, indicating where I would tap or swipe. During the process, I kept Nielsen’s heuristics in front of me and took detailed notes about interface issues, unclear elements, and navigation problems.

The information provided by the group was sufficient to understand the context and complete all tasks. The prototype itself had some unclear parts, which helped reveal several usability issues. Overall, the evaluation proceeded smoothly using the low-fidelity materials.

Part IV: List of Violations

1. H1 Visibility of system status
Where: Task 1 - Home screen
What: The Home screen does not indicate whether GPS/location services are active.
Why: Users depend on location feedback during hikes; lack of system status creates uncertainty about whether the system is ready.
Severity: 3
2. H1 Visibility of system status
Where: Task 3 - Confirmation screen
What: The confirmation message only says "Confirmation," without specifying that the tip was successfully added.
Why: Users need clear feedback about what action succeeded, especially in outdoor environments where actions may fail due to connectivity.
Severity: 2
3. H2 Match between system and the real world
Where: Task 3 - Confirmation screen
What: The actions after submitting ("Add Another" and "Return Home") do not match real hiking workflows. A hiker would more likely want to return to the route or map view, not the Home screen.
Why: This does not fit real-world expectations: hikers want to continue navigating the trail, not restart from the beginning.
Severity: 3
4. H4 Consistency and standards
Where: Task 3 - Confirmation screen
What: The button "Add Another" is vague; it does not specify "Add Another Tip," and may confuse users.
Why: Ambiguous terminology breaks consistency and forces the user to guess what the action does.
Severity: 2
5. H4 Consistency and standards
Where: All screens - smartwatch bezel buttons
What: The prototype includes four physical bezel buttons ("return," "next," "home," "custom"), which is uncommon for standard smartwatches.
Why: Most smartwatch platforms use 1-2 hardware buttons with consistent, predictable behaviors. Having four unlabeled, custom-function buttons breaks platform conventions and may confuse users.
Severity: 3

6. H6 Recognition rather than recall
Where: All screens - bezel buttons
What: The physical buttons have no labels, icons, or indicators explaining their function.
Why: Users must remember what each button does instead of recognizing it visually, increasing cognitive load during a hike.
Severity: 2
7. H2 Match between system and the real world
Where: Task 1 - Routes screen
What: When selecting a route, it is unclear whether the user is viewing details or actually “starting” the trail. There is no concept of “begin hike,” “start navigation,” or “follow this route.”
Why: In real hiking apps, selecting a route typically begins a navigation or overview mode. The lack of a clear start-of-journey state does not match user expectations for trail use.
Severity: 3
8. H2 Match between system and the real world
Where: Task 1 - Routes screen
What: The route list shows duration and distance but lacks key real-world hiking information such as elevation gain or trail difficulty rating.
Why: Elevation is one of the most important factors for hikers when choosing a trail. Missing this information makes the route selection incomplete.
Severity: 3
9. H3 User control and freedom
Where: Tip Details Popup
What: The popup window displays additional photos and descriptions but provides no visible way to close it or return to the previous screen.
Why: Users must always be able to undo or exit an unintended action, especially on small screens.
Severity: 4
10. H3 User control and freedom
Where: Task 2 - Tip List screen
What: The user must scroll through tips, but there is no “back to route overview” or “return to map” option visible. Only scrolling between tips is shown.
Why: Without a clear exit path, users may feel stuck in the tip list flow and unable to resume navigation quickly.
Severity: 3

11. H2 Match between system and the real world

Where: Task 2 - Tip List screen

What: The phrase "Current fork: Left is right" is ambiguous and can easily be misinterpreted.

Why: Trail guidance must reflect real-world clarity; unclear phrasing may cause hikers to take the wrong path.

Severity: 4 (Since it is Safety-related)

12. H7 Flexibility and efficiency of use

Where: Tip List screen

What: Users can only scroll through tips sequentially; there are no filters (e.g., by distance, danger level).

Why: This slows down experienced users and is inefficient in time-critical moments like reaching a confusing fork.

Severity: 2

13. H8 Aesthetic and minimalist design

Where: Task 1 - Routes screen

What: The Routes screen presents many icons (weather, danger, route duration, route shape) tightly packed into a very small circular display.

Why: The visual density makes it hard to read at a glance, which is crucial during hiking when users only glance at a smartwatch briefly.

Severity: 2

14. H9 Help users recognize, diagnose, and recover from errors

Where: Add Tip screen

What: If users try to submit an empty description or fail to choose an importance level, the system provides no error message explaining what to fix.

Why: Users receive no guidance on how to correct the issue, preventing successful task completion.

Severity: 3

15. HN Non-heuristic issue

Where: Task 2 - Tip List screen

What: The text content of tips varies greatly in tone and structure ("Watch out snakes-the green one is dangerous"), with no consistency guidelines.

Why: This is not a UI heuristic violation but a content-quality issue; crowd-generated content may require moderation or formatting rules for clarity.

Severity: N/A (since it is non-heuristic)

16. H5 Error prevention

Where: Task 3 - Add Tip screen

What: The interface allows submission with extremely short or unclear descriptions, and there is no indication of minimum input requirements.

Why: Preventing incomplete or low-quality tips is crucial for safety; the system should guide users before errors occur, not after.

Severity: 2

17. HN3 - Lack of Context for Photos / Media

Where: Tip Details popup (photo + text)

What: User-added photos in tips lack context such as timestamp, weather, or conditions.

Why: This is not a direct UI violation but a content completeness issue. For safety apps, knowing whether a photo is recent or outdated is crucial.

Severity: N/A

18. H10 - Help and documentation

Where: Route Details → “Add Tip” screen

What: The prototype does not provide any guidance on how to create a good or useful tip (e.g., what type of description to write, how long it should be, when to use each importance level).

Why: Since tip quality directly affects the usefulness of the app for other hikers, minimal in-interface guidance would help users understand what constitutes a helpful, complete contribution.

Severity: 2

Part V: Summary and Recommendations

Report in the table below the total number of identified violations.

Heuristic	# violations
H1: Visibility of system status	2
H2: Match between system and the real world	4
H3: User control and freedom	2
H4: Consistency and standards	2
H5: Error prevention	1
H6: Recognition rather than recall	1

H7: Flexibility and efficiency of use	1
H8: Aesthetic and minimalist design	1
H9: Help users recognize, diagnose, and recover from errors	1
H10: Help and documentation	1
HN: Non-heuristic issue	2

The TrailSight smartwatch prototype clearly communicates the main idea of helping hikers at confusing trail points. Working through the three assigned tasks made it easy to understand how the interface is meant to function during a hike. The overall flow between screens felt logical, and the group managed to fit the core features into the limited space of a smartwatch interface.

While using the prototype, I noticed some areas where the interaction could be clearer, especially around navigation, feedback, and how certain information is presented. Many of the issues relate to situations where hikers need quick, unambiguous guidance, and the current screens sometimes leave room for confusion. These are mostly practical refinements rather than major structural problems.

As a final note, I can see the smartwatch version working best together with the mobile prototype the group mentioned. Some actions—like writing longer descriptions or uploading photos—are more naturally handled on a phone, while the watch can focus on quick access during the hike. Improving clarity in a few key places, adding small pieces of guidance, and adjusting the navigation flow would make the smartwatch interface feel more confident and easier to use on the trail.